The contribution is a trace-to-frontier evidence contract

Inherited ingredients are explicit; the integration and full-corpus denominator are the contribution.

01 OBSERVED EVIDENCE

what the recorder actually knows

dynamic_trace.pb

PC • masks • lane sectors • placement

SASS control sidecar

stall/yield • barriers • wait masks • forms

architecture card

geometry • latency • typed assumptions

Identity-preserving join

function + PC + warp + launch

02 CAUSAL REPLAY

what the simulator contributes

1. **canonicalize**
   warp records + static control
2. **issue + wake**
   4 SMSP partitions • dependencies
3. **sector hierarchy**
   L1 → shared L2 → service envelope
4. **frontier ledger**
   issue ≠ completion ≠ observation ≠ tail

deterministic next-event order

observation-only

03 INDEPENDENT AUTHORITY

what is measured after replay

NCU profile directory

same capture identity • renamed kernel safe

launch-order fallback

98 exact + 11,460 ordered SM120 joins

metric decision

≤15% headline • >15% future work

Novelty boundary: not first SASS tracing • not first event simulation • not proprietary RTL recovery

Caption. Dynamic records, static controls, sector state, named frontiers, and independent NCU joins form one inspectable contract. Inherited mechanisms are labeled rather than rebranded as new.

Source: simulator crates, corrected 286/339 full-corpus joins, and SOTA audit in GPU_sims_notes.